

Sequence Flowcharts For Program Design

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What We'll Learn

By the end of this lesson, students can create a sequence flowchart that shows steps in the correct order from start to end.

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Words We'll Use

- Sequence
- Start
- End
- Input
- Process
- Output
- Flowline

Before We Start

Think about your morning routine before going to school.

What do you usually do first, next, and last?

Example:

1. Wake up
2. Take a bath
3. Wear uniform
4. Eat breakfast
5. Go to school

This kind of step-by-step order is called a sequence.

Let's Understand

A sequence flowchart shows instructions that happen one after another.

In programming, sequence means the computer follows each step in order. The first instruction is done first, the second instruction is done next, and so on until the program ends.

A basic sequence flowchart usually has:

- Start symbol
- Input symbol
- Process symbol
- Output symbol
- End symbol

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Example problem:

Create a flowchart that asks for a student's name and displays a greeting.

Problem analysis:

- Input: student name
- Process: none
- Output: greeting message

Flow:

Start → Input name → Display greeting → End

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Let's Practice

Create a sequence flowchart for each problem.

1. Ask for the user's name and display it.
2. Ask for two quiz scores and compute the total.
3. Ask for the price of an item and display the price.
4. Ask for the length and width of a rectangle, compute the area, and display the area.

Use the correct flowchart symbols for:

- Start and End
- Input
- Process
- Output
- Flowlines

Now We Apply

Create a sequence flowchart for this problem:

A program asks for three quiz scores, computes the average, and displays the average.

Problem analysis:

- Input: quiz1, quiz2, quiz3
- Process: $\text{average} = (\text{quiz1} + \text{quiz2} + \text{quiz3}) / 3$
- Output: Display average

Your flowchart should show the correct order of steps from Start to End.

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Challenge Task

Improve your flowchart by adding clear labels and a short explanation.

Your explanation should answer:

1. What inputs are needed?
2. What process is performed?
3. What output will be displayed?
4. Why is the order of steps important?

How Our Work Will Be Checked

- Correct use of flowchart symbols
- Steps are arranged in the correct sequence
- Flowlines clearly show the direction of the process
- Input, process, and output are complete
- Explanation is clear and easy to understand

Let's Reflect

Answer the following questions:

1. Why is sequence important in programming?
2. What happens if the steps in a flowchart are placed in the wrong order?
3. Which part of creating the flowchart was easiest for you?
4. Which part do you still need to practice?

Resources / References

- Flowchart symbol guide
- Sequence flowchart examples
- [images/sequence-flowchart.svg](#)
- [images/basic-sequence-symbols.svg](#)
- [images/greeting-flowchart.svg](#)
- [images/display-message-flowchart.svg](#)
- [images/add-two-numbers-flowchart.svg](#)
- [images/quiz-average-flowchart-guide.svg](#)

Assessment

<https://example.com/sequence-flowchart-assessment>